CES-FL Results in Ding et al.'s Table Format

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Important comparison note. Layout follows Ding et al.'s Tables 1–3: Table 1 reports FF++ per attribute; Tables 2–3 group gender, race, and intersection metrics by test dataset. Ding's published rows and the CES-FL archived rows are shown together for reference. CES-FL thresholds are validation-calibrated to a 10% empirical validation-real FPR; Ding's paper operating point and exact test inventory have not been verified as identical. These tables are therefore not an apples-to-apples claim, and no cross-method bold/underline ranking is applied. CES-FL is seed 17 only. Missing cells are shown as em dashes, never zero.

Table 1: Intra-domain fairness and detection results on FF++ (Ding et al. Table 1 layout). Values are percentages.

Attribute	Method	Backbone	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	AUC↑
Gender	Ori	Xception	4.10	5.72	91.93	92.69
	DAG-FDD	Xception	1.82	4.65	94.87	96.72
	DAW-FDD	Xception	0.78	9.52	95.76	97.46
	PG-FDD	Xception	0.62	4.74	96.32	97.66
	FairAdapter	ViT-L/14	4.16	12.21	67.85	71.50
	RSEF-FDD	Xception	0.57	8.55	94.91	97.09
	Ding (paper)	Xception	0.53	3.61	96.45	97.71
	CES-FL R2 (partial + DRO)†	Xception	0.88	3.85	97.02	98.40
Race	Ori	Xception	19.76	4.74	82.85	92.69
	DAG-FDD	Xception	5.48	9.20	93.43	96.72
	DAW-FDD	Xception	5.43	14.69	94.15	97.46
	PG-FDD	Xception	11.13	4.78	94.52	97.66
	FairAdapter	ViT-L/14	43.22	20.39	56.03	71.50
	RSEF-FDD	Xception	8.39	5.28	93.60	97.09
	Ding (paper)	Xception	9.29	4.35	94.86	97.71
	CES-FL R2 (partial + DRO)†	Xception	15.63	3.88	96.07	98.40
Intersection	Ori	Xception	36.03	14.64	74.43	92.69
	DAG-FDD	Xception	24.08	26.25	85.80	96.72
	DAW-FDD	Xception	14.36	26.13	86.74	97.46
	PG-FDD	Xception	9.19	13.39	86.83	97.66
	FairAdapter	ViT-L/14	86.91	42.44	45.98	71.50
	RSEF-FDD	Xception	23.64	21.74	85.77	97.09
	Ding (paper)	Xception	20.18	9.47	86.91	97.71
	CES-FL R2 (partial + DRO)†	Xception	30.23	12.26	90.57	98.40

Table 2: Cross-domain results on DFDC, Celeb-DF, and DFD (Ding et al. Table 2 layout). Values are percentages.

Dataset	Method	Backbone	Gender			Race			Intersection			Overall
			$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	AUC↑
DFDC	Ori	Xception	8.67	6.70	54.56	19.41	7.99	47.02	53.42	17.34	36.96	56.13
	DAG-FDD	Xception	5.29	6.68	57.41	9.26	12.51	45.91	45.30	12.92	36.42	59.13
	DAW-FDD	Xception	3.60	3.67	55.88	22.36	9.72	47.77	46.80	12.69	36.63	56.90
	PG-FDD	Xception	2.35	2.57	57.71	10.10	11.49	50.53	32.16	17.71	37.42	59.64
	FairAdapter	ViT-L/14	2.39	4.57	55.88	32.49	26.77	52.27	74.10	31.53	31.39	59.25
	RSEF-FDD	Xception	2.09	3.83	57.90	12.24	7.74	50.27	38.89	12.18	37.72	59.08
	Ding (paper)	Xception	1.76	3.67	57.94	17.93	7.58	52.33	37.37	11.92	39.03	61.86
	CES-FL R2 (partial + DRO)†	Xception	6.68	7.88	58.63	22.18	8.04	50.24	41.53	14.52	42.16	59.52
Celeb-DF	Ori	Xception	6.93	20.23	60.94	23.44	17.89	61.21	27.38	18.72	61.22	69.18
	DAG-FDD	Xception	8.70	14.25	66.23	19.44	13.26	62.61	23.60	13.80	63.08	72.29
	DAW-FDD	Xception	8.71	12.73	68.31	14.76	7.80	66.47	24.75	13.78	59.24	74.09
	PG-FDD	Xception	7.95	16.29	69.47	23.99	12.47	67.78	23.71	13.30	62.63	74.24
	FairAdapter	ViT-L/14	8.59	26.84	56.84	46.14	36.35	58.34	47.03	35.69	54.77	64.94
	RSEF-FDD	Xception	1.94	35.63	68.92	17.40	27.48	67.80	24.17	27.73	62.56	74.36
	Ding (paper)	Xception	6.41	10.81	69.68	16.60	7.29	69.96	23.32	12.01	63.27	76.75
	CES-FL R2 (partial + DRO)†	Xception	9.18	23.61	64.48	31.14	23.00	51.26	34.41	25.23	51.11	74.67
DFD	Ori	Xception	9.45	6.63	72.68	7.75	22.31	69.07	35.06	25.53	59.20	74.32
	DAG-FDD	Xception	5.41	5.60	75.87	2.56	21.34	70.14	33.00	23.02	61.09	76.29
	DAW-FDD	Xception	5.30	10.63	71.68	4.32	20.16	63.24	33.81	24.72	56.01	73.71
	PG-FDD	Xception	5.05	1.86	76.37	5.79	19.31	74.81	28.00	23.93	66.32	80.70
	FairAdapter	ViT-L/14	6.32	7.55	63.28	11.29	12.28	56.28	29.56	33.11	48.63	68.12
	RSEF-FDD	Xception	18.50	6.11	75.97	19.45	19.40	72.65	35.34	10.97	67.24	80.54
	Ding (paper)	Xception	4.72	5.49	78.98	2.09	18.66	77.44	27.32	22.81	69.73	81.46
	CES-FL R2 (partial + DRO)†	Xception	5.96	2.04	81.91	9.91	0.66	81.14	43.29	4.76	73.28	82.12

Table 3: Backbone comparison across FF++, DFDC, Celeb-DF, and DFD (Ding et al. Table 3 layout). Values are percentages.

Dataset	Method	Backbone	Gender			Race			Intersection			Overall
			$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	$es\text{-}AUC\uparrow$	AUC↑
FF++	Ori	ResNet-50	8.90	9.27	90.40	17.49	7.04	89.17	38.26	15.43	76.14	94.29
	DAG-FDD	ResNet-50	1.92	6.02	91.09	6.38	3.98	91.39	32.58	11.06	80.64	94.74
	DAW-FDD	ResNet-50	3.25	5.29	90.13	12.46	3.29	90.92	31.07	11.03	80.98	93.96
	PG-FDD	ResNet-50	3.05	3.06	93.57	10.89	2.83	91.53	22.10	9.69	81.58	94.46
	Ding (paper)	ResNet-50	3.94	5.14	94.46	18.67	2.43	92.69	34.30	8.61	82.87	95.53
	CES-FL R2 (partial + DRO)†	Xception	0.88	3.85	97.02	15.63	3.88	96.07	30.23	12.26	90.57	98.40
DFDC	Ori	ResNet-50	12.55	11.06	54.96	17.34	8.38	48.30	48.10	16.30	35.76	57.83
	DAG-FDD	ResNet-50	6.57	4.87	57.63	15.96	2.95	50.31	44.10	11.25	33.41	60.29
	DAW-FDD	ResNet-50	9.12	7.14	57.22	14.60	2.81	48.94	43.31	9.55	34.71	59.78
	PG-FDD	ResNet-50	5.78	4.65	58.34	12.68	5.23	49.33	31.97	8.54	34.85	61.83
	Ding (paper)	ResNet-50	3.97	3.61	56.96	11.94	5.15	50.99	32.14	10.27	37.55	62.54
	CES-FL R2 (partial + DRO)†	Xception	6.68	7.88	58.63	22.18	8.04	50.24	41.53	14.52	42.16	59.52
Celeb-DF	Ori	ResNet-50	12.26	22.12	61.11	18.55	22.43	59.82	24.84	23.18	56.09	71.65
	DAG-FDD	ResNet-50	8.22	20.45	67.01	23.07	18.50	68.07	20.29	19.33	66.76	75.77
	DAW-FDD	ResNet-50	9.03	20.85	68.33	18.60	17.65	69.14	23.84	18.55	68.74	76.51
	PG-FDD	ResNet-50	6.98	16.92	68.49	16.74	16.50	70.35	29.34	18.72	66.26	76.10
	Ding (paper)	ResNet-50	4.53	12.00	68.98	6.76	8.16	71.03	5.56	9.11	68.25	77.35
	CES-FL R2 (partial + DRO)†	Xception	9.18	23.61	64.48	31.14	23.00	51.26	34.41	25.23	51.11	74.67
DFD	Ori	ResNet-50	10.95	9.08	70.33	4.66	12.57	66.68	33.46	16.84	56.00	71.31
	DAG-FDD	ResNet-50	3.62	4.19	73.74	3.86	13.54	70.33	25.22	14.84	57.19	73.97
	DAW-FDD	ResNet-50	4.81	3.90	72.12	7.36	3.21	70.57	33.45	14.30	54.52	73.58
	PG-FDD	ResNet-50	3.57	1.82	76.14	2.62	7.81	72.16	23.15	11.01	62.65	77.47
	Ding (paper)	ResNet-50	3.12	1.76	77.68	6.51	3.03	70.93	20.99	4.85	66.19	78.63
	CES-FL R2 (partial + DRO)†	Xception	5.96	2.04	81.91	9.91	0.66	81.14	43.29	4.76	73.28	82.12

† CES-FL R2 is the single Xception row shown: partial fine-tuning + DRO, seed 17. It has the highest arithmetic mean AUC (78.68%) across the four archived Xception test sets among the listed CES-FL arms. This is a post-hoc descriptive choice, not a preregistered model selection; the older ResNet B/C pilots are omitted. Table 3 places this Xception row after Ding et al.'s ResNet-50 rows; the backbone difference is explicit. **Comparison limitation:** Ding et al. report 50 epochs and their own paper operating point. The CES-FL values here are single-seed archived results and were not proven to use an identical test inventory, threshold, or training schedule. Do not interpret this layout as a matched benchmark or claim a general Ding win.

Forty-metric summary

For the selected CES-FL R2 reference, 14 of 40 point estimates are numerically favorable to Ding under the preferred direction. This is descriptive only: fairness thresholds and test inventories are not confirmed to match. R2 AUC is above Ding's reported AUC on FF++ (98.40 vs 97.71), and DFD (82.12 vs 81.46), and below on DFDC (59.52 vs 61.86) and Celeb-DF (74.67 vs 76.75). No multi-seed uncertainty estimates are available.

Table 4: FairForge-2 comparison with Ding-style fairness metrics. Values are percentages; full and public splits are separated from FF++-trained transfer evaluations.

Protocol	Model	Gender			Race			Intersection			Overall	Test
		$F_{FPR}\downarrow$	$F_{DP}\downarrow$	*es-AUC* $\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	*es-AUC* $\uparrow$	$F_{FPR}\downarrow$	$F_{DP}\downarrow$	*es-AUC* $\uparrow$	AUC$\uparrow$	*N*
FF2 internal	Ding classification	—	—	—	—	—	—	—	—	—	99.2619	182884
FF2 internal	Ding SFD+GDA	—	—	—	—	—	—	—	—	—	99.3573	182884
FF2 internal	Older CFES detection-only	—	—	—	—	—	—	—	—	—	97.6043	182884
FF2 internal	Older CFES fairness run	—	—	—	—	—	—	—	—	—	98.0969	182884
FF++ → FF2	Ding Ori, Xception seed 7	3.23	1.06	41.75	25.27	17.64	39.76	69.79	29.76	36.26	41.85	182884
FF++ → FF2	Ding SFD+GDA, Xception seed 7	3.00	3.93	42.29	30.01	21.39	40.64	79.58	30.24	37.91	42.40	182884
FF2 public	Older CFES-D	—	—	—	—	—	—	—	—	—	98.2598	87564
FF2 public	Current CES-FL R2	—	—	—	—	—	—	—	—	—	—	Not run

The FF++→FF2 rows are separate transfer evaluations from archived FF++ checkpoints. Their group statistics use the evaluator's fixed probability cutoff 0.5 and Ding Eqs. 8–10; their low AUCs (41.85%, 42.40%) must not be confused with the FF2-trained in-domain checkpoints above. Fairness metrics for the four in-domain full-split rows and the public CFES-D row were not available for their matching saved-score/checkpoint artifacts, so they are left blank. An FF2-trained current CES-FL run does not exist in the audited results.

Internal/full manifests: 502,035 TRAIN / 74,974 SELECT / 182,884 TEST. Public manifests: 218,266 TRAIN / 31,107 SELECT / 87,564 TEST. The internal Ding run was a local budget-limited 5-epoch classification + 5-epoch GDA archive, not a verified exact paper reproduction.

The FF++→FF2 rows are separate transfer evaluations from archived FF++ checkpoints. Their group statistics use the evaluator's fixed probability cutoff 0.5 and Ding Eqs. 8–10; their low AUCs (41.85%, 42.40%) must not be confused with the FF2-trained in-domain checkpoints above. Fairness metrics for the four in-domain full-split rows and the public CFES-D row were not available for their matching saved-score/checkpoint artifacts, so they are left blank. An FF2-trained current CES-FL run does not exist in the audited results. Internal/full manifests: 502,035 TRAIN / 74,974 SELECT / 182,884 TEST. Public manifests: 218,266 TRAIN / 31,107 SELECT / 87,564 TEST. The internal Ding run was a local budget-limited 5-epoch classification + 5-epoch GDA archive, not a verified exact paper reproduction.

Sources

Ding et al., Tables 1–3 of arXiv:2511.10150v5. Archived CES-FL comparison rows: commit 39471dc, accompanying CSV. FairForge-2 results audit: historical FF2 audit. Ding published values are transcribed from the paper tables. CES-FL extracted rows are rounded here to two decimals; missing values are retained as em dashes.